

Grade 7
Unit 6
Lesson 6 Practice

Name _____
Date _____
Period _____

A map is set to a scale factor of 37 mm = 1 km. Use this scale to answer questions 1-3.

1. *Two* mountains are 351.1 mm apart on the map. Create a proportion that could be used to determine the actual distance between the mountains.

$$\frac{37 \text{ mm}}{1 \text{ km}} = \frac{351.1 \text{ mm}}{x}$$

$$\frac{37x}{37} = \frac{351.1}{37}$$

$$x = 9.49 \text{ km}$$

2. Use proportions to solve for the actual distance between the mountains. Round the answer to the nearest tenth.

$$\frac{37x}{37} = \frac{351.1}{37}$$

$$x = 9.5 \text{ km}$$

3. Using the scale from the map in question 1, if the distance between *two* towns is 73 kilometers, how many millimeters apart will the towns be on the map?

$$\frac{37 \text{ mm}}{1 \text{ km}} = \frac{x}{73}$$

$$x = 2701 \text{ mm}$$

In a construction zone, there are black and gray beams that are being used for a building that are proportional in size. The black beams are 6 feet wide and 18 feet long, and the gray beams are 11.5 ft long. Use this information to answer questions 4-5.

4. Fill in the remaining values for the proportion below.

$$\frac{6}{x} = \frac{18}{11.5}$$

5. Using the above proportion, determine the width of the gray beams.

$$\frac{18x}{18} = \frac{69}{18}$$

$$x = 3.8 \text{ ft}$$

Use the following information to solve problems 6-10. Suppose 8,900 people live in apartment buildings in Charlotte County, Florida. A sample of 320 people were polled asking them, if given the opportunity, would they move into a new house.

- 147 said yes.
- 155 said no.
- 18 were undecided.

6. Create a proportion to determine the total number of people who were undecided.

$$\frac{18}{320} = \frac{x}{8900}$$

7. Create a proportion to determine the total number of people who said yes.

$$\frac{147}{320} = \frac{x}{8900}$$

8. Create a proportion to determine the total number of people who said no.

$$\frac{155}{320} = \frac{x}{8900}$$

9. To the nearest whole number, determine how many people were undecided.

$$\frac{18}{320} = \frac{x}{8900}$$

$$\frac{320x}{320} = \frac{160,200}{320}$$

$$x \approx 500.6$$

$$x = 501$$

10. To the nearest whole number, determine the number of people who said yes.

$$\frac{147}{320} = \frac{x}{8900}$$

$$\frac{320x}{320} = \frac{1,308,300}{320}$$

$$x \approx 4088.4$$

$$x = 4,088$$